

Deena Dhayalan

Software Engineer | Big Data Engineer

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SUMMARY

Engineer with hands-on experience designing and implementing high-performance data pipelines using modern big data technologies. Proficient in working with distributed systems such as Kafka, Hadoop, Hive, Trino, and Apache Pinot to enable scalable data ingestion, transformation, and querying. Skilled in building reliable, low-latency pipelines for both batch and real-time workloads. Experienced in optimizing query performance and managing large-scale data infrastructure that powers analytics and business decision-making.

SKILLS

- **Programming Languages:** Python, Java, SQL
- **Backend & Architecture:** Distributed Systems, System design, Event-driven Architecture & APIs
- **AI / Machine Learning:** Generative AI (LLMs), Prompt Engineering, LangChain, LangGraph, RAG systems, Semantic Search, Agentic Reasoning, and Model Evaluation.
- **Data Engineering:** Data Modeling, ETL/ELT, Data Warehouse, Data LakeHouse, SQL Optimization
- **Databases:** MySQL, PostgreSQL
- **Tools & Platforms:** Hadoop, Hive, Kafka, Pinot, Trino, ClickHouse, Redis, AWS S3, Ant & Gradle
- **DevOps & Platform:** Docker, Kubernetes, CI/CD (GitHub Actions)

EXPERIENCE

- **Zoho Corporation – Software Engineer** Chennai, India
Full-time July 2022 – Present
 - **BDaaS (Big Data as a Service)**
 - Ensured secure and scalable deployment of AI capabilities within a global SaaS platform.
 - Built a high-performance ingestion service/CDC to transport data from MySQL/PostgreSQL to a data lakehouse capable of handling 5TB/day.
 - Developed an admin service to onboard and manage internal Zoho services. Successfully onboarded major services like Zoho CRM and ManageEngine.
 - Established and integrated Apache tools such as Pinot, Kafka, HDFS, Hive, and Trino into BDaaS systems.
 - Developed low-latency processing pipelines using Kafka Consumer APIs and Kafka Streams capable of handling up to 5 billion rows per hour.
 - Integrated query engines like Apache Pinot and Trino to enable low-latency querying.
 - Built a data storage layer using Apache Hadoop following medallion architecture. Utilized ORC file format for better compression and faster retrieval.
 - Mentored a team of junior engineers in distributed systems design and clean code principles, improving team delivery velocity and code quality through rigorous peer reviews
- **Zoho Corporation – Project Intern** Chennai, India
Internship Jan 2022 – June 2022
 - **Internship Experience**
 - Contributed to the WebAnalytics platform by building backend modules to process and analyze user interaction data.
 - Conducted a POC on Apache Pinot for sub-second analytics over high-volume streaming data, achieving over 2 million events/hour in stress tests.
 - Gained exposure to distributed systems, low-latency streaming, and real-time analytics infrastructure.

PROJECTS

- **Data Viewer with Trino:** Designed a unified data exploration tool that aggregates datasets from multiple internal platforms. Facilitated seamless ad hoc querying over near real-time data using Trino, reducing reporting overhead and accelerating data-driven decisions.
- **Multipart Upload System:** Developed a backend service that supports multi-part file uploads for large payloads. The system ensures chunk validation, ordering, and automatic compaction into final files, improving reliability and scalability of large file handling.

EDUCATION

- **Kumaraguru College of Technology** Tamil Nadu, India
B.Tech. in Information Science and Engineering; GPA: 9.1 July 2018 – April 2022

PUBLICATIONS

- **Agriculture-Based Recommendation System with Image Processing:** Published in IEEE. The system recommends suitable crops and predicts plant diseases using live soil and weather data combined with image analysis. [\[IEEE Link\]](#)

ACHIEVEMENTS

- **Chess Champion:** Won the Zonal-level Chess Tournament.
- **State Chess Ranking:** Represented TamilNadu State Chess Championships.